Always model your data so that each index has one particular data shape. When we index a document, we create a one-to-one mapping between the document and the index. That is, one index can only have one document type. We follow this principle throughout the book.

Now that we know our data is presented as JSON documents to be stored in Elasticsearch, the next logical step is to find out where these documents are stored. Just like a table in a database that hosts records, a special bucket (collection) called an *index* holds all documents of a particular shape in Elasticsearch. Details are in the next section.

3.2.2 Indexes

We need a container to host our documents in the store. For this, Elasticsearch creates an index as a logical collection of documents. Just as we keep paper documents in a filing cabinet, Elasticsearch keeps data documents in an index, except the index is not a physical storage place; it is a logical grouping. It is composed of (or backed up by) *shards*. Figure 3.5 shows a `cars` index composed of three shards on three nodes (a *node* is an instance of Elasticsearch), one shard per node.

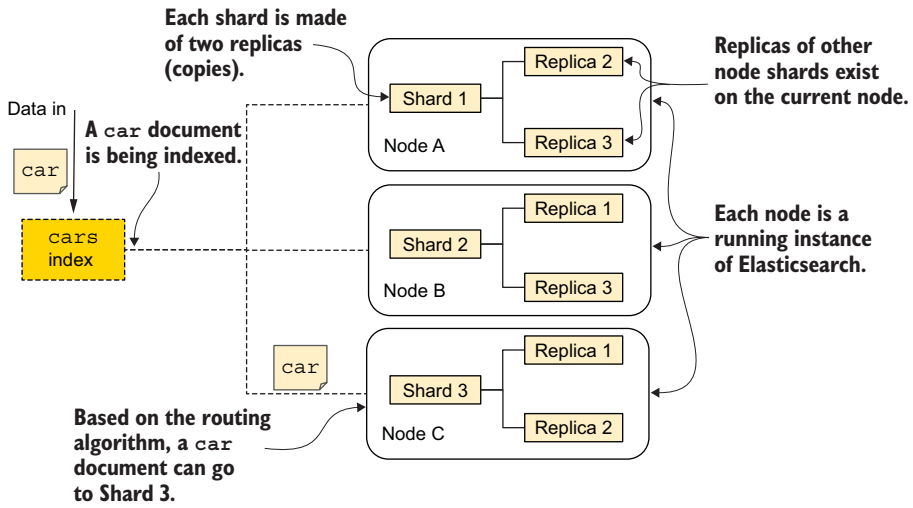


Figure 3.5 An index designed with three shards and two replicas per shard

In addition to having three shards, the index is declared to have *two replicas per shard*, both hosted on other nodes. That leads to a question: what is a shard?

Shards are the physical instances of Apache Lucene, the workhorses behind the scenes in getting our data in and out of storage. In other words, shards take care of the physical storage and retrieval of our data.